

# Power BI – DAX Logical Functions

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Read

Data analysis expressions, or DAX, are formulas or expressions used in calculations and data analysis. These expressions contain a variety of different functions, operators, and constants that are all evaluated collectively as a single formula to yield the desired outcomes (value or values). DAX formulas are very beneficial in BI tools like Power BI because they help data analysts get the most out of the data sets they already have.

## DAX Functions

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A DAX function is a previously written formula that performs calculations using input values. A column reference, numbers, text, constants, references to other formulas or functions, and logical values like TRUE or FALSE are all acceptable function arguments. They must be supplied to the function in a particular sequence. Every function performs a particular operation on the values in an argument. Multiple arguments can be supported by a DAX formula.

## Types of DAX Functions

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- Date and Time Functions
- Time Intelligence Functions
- Information Functions
- Logical Functions
- Mathematical and Trigonometric Functions
- Statistical Functions
- Text Functions
- Parent-Child functions
- Other functions
- Table functions

The different DAX function types are described above. A DAX function is a previously created formula that computes using values passed to it as arguments. We now look at DAX Logical Functions:

## DAX Logical Functions

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The Logical function of Power BI is among its most crucial components. It offers up a result as either True or False. Logical functions return information about the values or sets in an expression. For instance, you can check an expression's output and produce conditional results using the IF function. In order to study these functions, you may look at the [dataset](#). Below is the screenshot of the dataset:

| Country                  | Product | Qty Sold | Year of Sales | Gross Sales | Net Sales |
|--------------------------|---------|----------|---------------|-------------|-----------|
| Canada                   | Pen     | 20       | 2019          | 420         | 400       |
| Germany                  | Mouse   | 15       | 2019          | 30          | 15        |
| France                   | CPU     | 1999     | 2018          | 9995        | 5997      |
| Germany                  | Pen     | 350      | 2019          | 2450        | 350       |
| Mexico                   | CPU     | 12       | 2018          | 156         | 96        |
| Germany                  | Mouse   | 7        | 2020          | 84          | 70        |
| Germany                  | Pencil  | 15       | 2019          | 45          | 15        |
| Canada                   | Pencil  | 120      | 2020          | 5280        | 4800      |
| France                   | Pen     | 15       | 2019          | 315         | 165       |
| Germany                  | CPU     | 7        | 2018          | 238         | 140       |
| Mexico                   | Mouse   | 22       | 2020          | 462         | 242       |
| Canada                   | Mouse   | 12       | 2019          | 672         | 276       |
| Mexico                   | Mouse   | 15       | 2019          | 825         | 330       |
| Germany                  | Mouse   | 20       | 2019          | 1540        | 660       |
| Canada                   | Pen     | 800      | 2018          | 71200       | 64800     |
| United States of America | Pen     | 7        | 2019          | 63          | 21        |
| Canada                   | Pen     | 125      | 2020          | 9500        | 9375      |

## IF

A condition is checked, and if it is TRUE, one value is returned; otherwise, a different value is returned.

**Syntax:** IF(<logical\_test>, <value\_if\_true>[, <value\_if\_false>])

**Example:**

| 1 Sales_Status = IF('Sales Summary'[Net Sales]>500,"High","Low") |         |          |               |             |           |              |
|------------------------------------------------------------------|---------|----------|---------------|-------------|-----------|--------------|
| Country                                                          | Product | Qty Sold | Year of Sales | Gross Sales | Net Sales | Sales_Status |
| Canada                                                           | Pen     | 20       | 2019          | 420         | 400       | Low          |
| Germany                                                          | Mouse   | 15       | 2019          | 30          | 15        | Low          |
| France                                                           | CPU     | 1999     | 2018          | 9995        | 5997      | High         |
| Germany                                                          | Pen     | 350      | 2019          | 2450        | 350       | Low          |
| Mexico                                                           | CPU     | 12       | 2018          | 156         | 96        | Low          |
| Germany                                                          | Mouse   | 7        | 2020          | 84          | 70        | Low          |
| Germany                                                          | Pencil  | 15       | 2019          | 45          | 15        | Low          |
| Canada                                                           | Pencil  | 120      | 2020          | 5280        | 4800      | High         |
| France                                                           | Pen     | 15       | 2019          | 315         | 165       | Low          |
| Germany                                                          | CPU     | 7        | 2018          | 238         | 140       | Low          |
| Mexico                                                           | Mouse   | 22       | 2020          | 462         | 242       | Low          |
| Canada                                                           | Mouse   | 12       | 2019          | 672         | 276       | Low          |
| Mexico                                                           | Mouse   | 15       | 2019          | 825         | 330       | Low          |
| Germany                                                          | Mouse   | 20       | 2019          | 1540        | 660       | High         |
| Canada                                                           | Pen     | 800      | 2018          | 71200       | 64800     | High         |
| United States of America                                         | Pen     | 7        | 2019          | 63          | 21        | Low          |
| Canada                                                           | Pen     | 125      | 2020          | 9500        | 9375      | High         |

## AND

The function determines whether both arguments are TRUE and returns TRUE if they are. If not, returns false.

**Syntax:** AND(<logical1>,<logical2>)

**Example:**

| X ✓                      |         | 1 CanadaSales = IF(AND('Sales Summary'[Country]="Canada",'Sales Summary'[Net Sales]>500),"Valid","Invalid") |               |             |           |             |
|--------------------------|---------|-------------------------------------------------------------------------------------------------------------|---------------|-------------|-----------|-------------|
| Country                  | Product | Qty Sold                                                                                                    | Year of Sales | Gross Sales | Net Sales | CanadaSales |
| Canada                   | Pen     | 20                                                                                                          | 2019          | 420         | 400       | Invalid     |
| Germany                  | Mouse   | 15                                                                                                          | 2019          | 30          | 15        | Invalid     |
| France                   | CPU     | 1999                                                                                                        | 2018          | 9995        | 5997      | Invalid     |
| Germany                  | Pen     | 350                                                                                                         | 2019          | 2450        | 350       | Invalid     |
| Mexico                   | CPU     | 12                                                                                                          | 2018          | 156         | 96        | Invalid     |
| Germany                  | Mouse   | 7                                                                                                           | 2020          | 84          | 70        | Invalid     |
| Germany                  | Pencil  | 15                                                                                                          | 2019          | 45          | 15        | Invalid     |
| Canada                   | Pencil  | 120                                                                                                         | 2020          | 5280        | 4800      | Valid       |
| France                   | Pen     | 15                                                                                                          | 2019          | 315         | 165       | Invalid     |
| Germany                  | CPU     | 7                                                                                                           | 2018          | 238         | 140       | Invalid     |
| Mexico                   | Mouse   | 22                                                                                                          | 2020          | 462         | 242       | Invalid     |
| Canada                   | Mouse   | 12                                                                                                          | 2019          | 672         | 276       | Invalid     |
| Mexico                   | Mouse   | 15                                                                                                          | 2019          | 825         | 330       | Invalid     |
| Germany                  | Mouse   | 20                                                                                                          | 2019          | 1540        | 660       | Invalid     |
| Canada                   | Pen     | 800                                                                                                         | 2018          | 71200       | 64800     | Valid       |
| United States of America | Pen     | 7                                                                                                           | 2019          | 63          | 21        | Invalid     |
| Canada                   | Pen     | 125                                                                                                         | 2020          | 9500        | 9375      | Valid       |

## COALESCE

Identifies the first expression that is not BLANK and returns it. BLANK is returned if all expressions evaluate to NULL.

**Syntax:** COALESCE(<expression>, <expression>[, <expression>]...)

**Example:**

| 1 CoalDemo = COALESCE('Sales Summary'[CountryCode],BLANK(),BLANK(),"Default",BLANK()) |         |          |               |             |           |         |             |          |         |
|---------------------------------------------------------------------------------------|---------|----------|---------------|-------------|-----------|---------|-------------|----------|---------|
| Country                                                                               | Product | Qty Sold | Year of Sales | Gross Sales | Net Sales | NotDemo | CountryCode | CoalDemo |         |
| Canada                                                                                | Pen     | 20       | 2019          | 420         | 400       | Invalid | CA          | CA       | CA      |
| Germany                                                                               | Mouse   | 15       | 2019          | 30          | 15        | Valid   | GE          | GE       | GE      |
| France                                                                                | CPU     | 1999     | 2018          | 9995        | 5997      | Valid   | FR          | FR       | FR      |
| Germany                                                                               | Pen     | 350      | 2019          | 2450        | 350       | Valid   | GE          | GE       | GE      |
| Mexico                                                                                | CPU     | 12       | 2018          | 156         | 96        | Valid   |             | Default  | Default |
| Germany                                                                               | Mouse   | 7        | 2020          | 84          | 70        | Valid   | GE          | GE       | GE      |
| Germany                                                                               | Pencil  | 15       | 2019          | 45          | 15        | Valid   | GE          | GE       | GE      |
| Canada                                                                                | Pencil  | 120      | 2020          | 5280        | 4800      | Invalid | CA          | CA       | CA      |
| France                                                                                | Pen     | 15       | 2019          | 315         | 165       | Valid   | FR          | FR       | FR      |
| Germany                                                                               | CPU     | 7        | 2018          | 238         | 140       | Valid   | GE          | GE       | GE      |
| Mexico                                                                                | Mouse   | 22       | 2020          | 462         | 242       | Valid   |             | Default  | Default |
| Canada                                                                                | Mouse   | 12       | 2019          | 672         | 276       | Invalid | CA          | CA       | CA      |
| Mexico                                                                                | Mouse   | 15       | 2019          | 825         | 330       | Valid   |             | Default  | Default |
| Germany                                                                               | Mouse   | 20       | 2019          | 1540        | 660       | Valid   | GE          | GE       | GE      |
| Canada                                                                                | Pen     | 800      | 2018          | 71200       | 64800     | Invalid | CA          | CA       | CA      |
| United States of America                                                              | Pen     | 7        | 2019          | 63          | 21        | Valid   |             | Default  | Default |
| Canada                                                                                | Pen     | 125      | 2020          | 9500        | 9375      | Invalid | CA          | CA       | CA      |

## IFERROR

Returns the value of the expression itself unless the expression returns an error, in which case it returns a specified value.

**Syntax:** IFERROR(value, value\_if\_error)

**Example:**

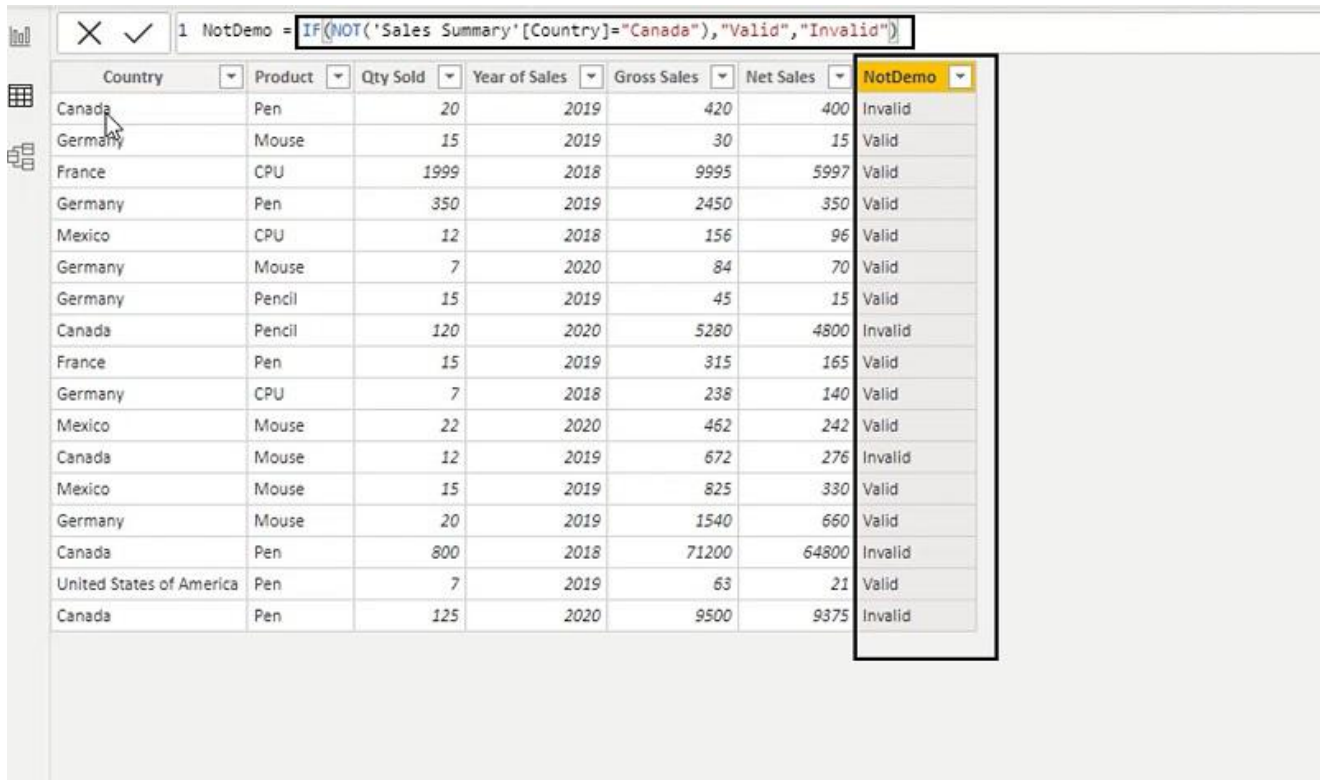
| 1 ErrorDemo = IFERROR('Sales Summary'[Net Sales],0) |         |          |               |             |           |         |             |           |       |
|-----------------------------------------------------|---------|----------|---------------|-------------|-----------|---------|-------------|-----------|-------|
| Country                                             | Product | Qty Sold | Year of Sales | Gross Sales | Net Sales | NotDemo | CountryCode | ErrorDemo |       |
| Canada                                              | Pen     | 20       | 2019          | 420         | 400       | Invalid | CA          | 400       | 400   |
| Germany                                             | Mouse   | 15       | 2019          | 30          | 15        | Valid   | GE          | 15        | 15    |
| France                                              | CPU     | 1999     | 2018          | 9995        | 5997      | Valid   | FR          | 5997      | 5997  |
| Germany                                             | Pen     | 350      | 2019          | 2450        | 350       | Valid   | GE          | 350       | 350   |
| Mexico                                              | CPU     | 12       | 2018          | 156         | 96        | Valid   |             | 96        | 96    |
| Germany                                             | Mouse   | 7        | 2020          | 84          | 70        | Valid   | GE          | 70        | 70    |
| Germany                                             | Pencil  | 15       | 2019          | 45          | 15        | Valid   | GE          | 15        | 15    |
| Canada                                              | Pencil  | 120      | 2020          | 5280        | 4800      | Invalid | CA          | 4800      | 4800  |
| France                                              | Pen     | 15       | 2019          | 315         | 165       | Valid   | FR          | 165       | 165   |
| Germany                                             | CPU     | 7        | 2018          | 238         | 140       | Valid   | GE          | 140       | 140   |
| Mexico                                              | Mouse   | 22       | 2020          | 462         | 242       | Valid   |             | 242       | 242   |
| Canada                                              | Mouse   | 12       | 2019          | 672         | 276       | Invalid | CA          | 276       | 276   |
| Mexico                                              | Mouse   | 15       | 2019          | 825         | 330       | Valid   |             | 330       | 330   |
| Germany                                             | Mouse   | 20       | 2019          | 1540        | 660       | Valid   | GE          | 660       | 660   |
| Canada                                              | Pen     | 800      | 2018          | 71200       | 64800     | Invalid | CA          | 64800     | 64800 |
| United States of America                            | Pen     | 7        | 2019          | 63          | 21        | Valid   |             | 21        | 21    |
| Canada                                              | Pen     | 125      | 2020          | 9500        | 9375      | Invalid | CA          | 9375      | 9375  |

## NOT

This function converts either FALSE or TRUE values into the other.

**Syntax:** NOT(<logical>)

**Example:**



The screenshot shows a data table with columns: Country, Product, Qty Sold, Year of Sales, Gross Sales, Net Sales, and NotDemo. The formula bar at the top displays the formula: `=IF(NOT('Sales Summary'[Country]="Canada"),"Valid","Invalid")`. The NotDemo column contains values 'Valid' or 'Invalid' based on whether the Country is 'Canada'.

| Country                  | Product | Qty Sold | Year of Sales | Gross Sales | Net Sales | NotDemo |
|--------------------------|---------|----------|---------------|-------------|-----------|---------|
| Canada                   | Pen     | 20       | 2019          | 420         | 400       | Invalid |
| Germany                  | Mouse   | 15       | 2019          | 30          | 15        | Valid   |
| France                   | CPU     | 1999     | 2018          | 9995        | 5997      | Valid   |
| Germany                  | Pen     | 350      | 2019          | 2450        | 350       | Valid   |
| Mexico                   | CPU     | 12       | 2018          | 156         | 96        | Valid   |
| Germany                  | Mouse   | 7        | 2020          | 84          | 70        | Valid   |
| Germany                  | Pencil  | 15       | 2019          | 45          | 15        | Valid   |
| Canada                   | Pencil  | 120      | 2020          | 5280        | 4800      | Invalid |
| France                   | Pen     | 15       | 2019          | 315         | 165       | Valid   |
| Germany                  | CPU     | 7        | 2018          | 238         | 140       | Valid   |
| Mexico                   | Mouse   | 22       | 2020          | 462         | 242       | Valid   |
| Canada                   | Mouse   | 12       | 2019          | 672         | 276       | Invalid |
| Mexico                   | Mouse   | 15       | 2019          | 825         | 330       | Valid   |
| Germany                  | Mouse   | 20       | 2019          | 1540        | 660       | Valid   |
| Canada                   | Pen     | 800      | 2018          | 71200       | 64800     | Invalid |
| United States of America | Pen     | 7        | 2019          | 63          | 21        | Valid   |
| Canada                   | Pen     | 125      | 2020          | 9500        | 9375      | Invalid |

## OR

Returns TRUE if one of the arguments is verified to be TRUE. If both arguments are false, the function returns FALSE.

**Syntax:** OR(<logical1>,<logical2>)

**Example:**

1 CanadaSales = IF(OR('Sales Summary'[Country]="Canada",'Sales Summary'[Net Sales]>500),"Valid","Invalid")

| Country                  | Product | Qty Sold | Year of Sales | Gross Sales | Net Sales | CanadaSales |
|--------------------------|---------|----------|---------------|-------------|-----------|-------------|
| Canada                   | Pen     | 20       | 2019          | 420         | 400       | Valid       |
| Germany                  | Mouse   | 15       | 2019          | 30          | 15        | Invalid     |
| France                   | CPU     | 1999     | 2018          | 9995        | 5997      | Valid       |
| Germany                  | Pen     | 350      | 2019          | 2450        | 350       | Invalid     |
| Mexico                   | CPU     | 12       | 2018          | 156         | 96        | Invalid     |
| Germany                  | Mouse   | 7        | 2020          | 84          | 70        | Invalid     |
| Germany                  | Pencil  | 15       | 2019          | 45          | 15        | Invalid     |
| Canada                   | Pencil  | 120      | 2020          | 5280        | 4800      | Valid       |
| France                   | Pen     | 15       | 2019          | 315         | 165       | Invalid     |
| Germany                  | CPU     | 7        | 2018          | 238         | 140       | Invalid     |
| Mexico                   | Mouse   | 22       | 2020          | 462         | 242       | Invalid     |
| Canada                   | Mouse   | 12       | 2019          | 672         | 276       | Valid       |
| Mexico                   | Mouse   | 15       | 2019          | 825         | 330       | Invalid     |
| Germany                  | Mouse   | 20       | 2019          | 1540        | 660       | Valid       |
| Canada                   | Pen     | 800      | 2018          | 71200       | 64800     | Valid       |
| United States of America | Pen     | 7        | 2019          | 63          | 21        | Invalid     |
| Canada                   | Pen     | 125      | 2020          | 9500        | 9375      | Valid       |

## SWITCH

Returns one of several potential result expressions after comparing an expression to a list of values.

**Syntax:** SWITCH(<expression>, <value>, <result>[, <value>, <result>]...[, <else>])

**Example:**

1 CountryCode = SWITCH('Sales Summary'[Country],"Canada","CA","France","FR","Germany","GE")

| Country                  | Product | Qty Sold | Year of Sales | Gross Sales | Net Sales | NotDemo | CountryCode |
|--------------------------|---------|----------|---------------|-------------|-----------|---------|-------------|
| Canada                   | Pen     | 20       | 2019          | 420         | 400       | Invalid | CA          |
| Germany                  | Mouse   | 15       | 2019          | 30          | 15        | Valid   | GE          |
| France                   | CPU     | 1999     | 2018          | 9995        | 5997      | Valid   | FR          |
| Germany                  | Pen     | 350      | 2019          | 2450        | 350       | Valid   | GE          |
| Mexico                   | CPU     | 12       | 2018          | 156         | 96        | Valid   |             |
| Germany                  | Mouse   | 7        | 2020          | 84          | 70        | Valid   | GE          |
| Germany                  | Pencil  | 15       | 2019          | 45          | 15        | Valid   | GE          |
| Canada                   | Pencil  | 120      | 2020          | 5280        | 4800      | Invalid | CA          |
| France                   | Pen     | 15       | 2019          | 315         | 165       | Valid   | FR          |
| Germany                  | CPU     | 7        | 2018          | 238         | 140       | Valid   | GE          |
| Mexico                   | Mouse   | 22       | 2020          | 462         | 242       | Valid   |             |
| Canada                   | Mouse   | 12       | 2019          | 672         | 276       | Invalid | CA          |
| Mexico                   | Mouse   | 15       | 2019          | 825         | 330       | Valid   |             |
| Germany                  | Mouse   | 20       | 2019          | 1540        | 660       | Valid   | GE          |
| Canada                   | Pen     | 800      | 2018          | 71200       | 64800     | Invalid | CA          |
| United States of America | Pen     | 7        | 2019          | 63          | 21        | Valid   |             |
| Canada                   | Pen     | 125      | 2020          | 9500        | 9375      | Invalid | CA          |

Below is the List of Logical DAX Functions in PowerBI:

| Function         | Description                                                                                                                                                                                                                      | Syntax                                                        |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>IF</b>        | A condition is checked, and if it is TRUE, one value is returned; otherwise, a different value is returned.                                                                                                                      | IF(<logical_test>, <value_if_true>[, <value_if_false>])       |
| <b>IF.EAGER</b>  | If the condition is true, the check returns one value; if not, a different value is returned. The branch expressions are always executed regardless of the condition expression thanks to the eager execution plan that is used. | IF.EAGER(<logical_test>, <value_if_true>[, <value_if_false>]) |
| <b>IFERROR</b>   | Returns the value of the expression itself unless the expression returns an error, in which case it returns a specified value.                                                                                                   | IFERROR(value, value_if_error)                                |
| <b>AND</b>       | The function determines whether both arguments are TRUE and returns TRUE if they are. If not, returns false.                                                                                                                     | AND(<logical1>, <logical2>)                                   |
| <b>BITAND</b>    | The output of this function is the bitwise AND of two numbers.                                                                                                                                                                   | BITAND(<number>, <number>)                                    |
| <b>BITLSHIFT</b> | Returns a number with the specified number of bits shifted to the left.                                                                                                                                                          | BITLSHIFT(<Number>, <Shift_Amount>)                           |
| <b>BITOR</b>     | It provides a bitwise OR of two numbers as a result.                                                                                                                                                                             | BITOR(<number>, <number>)                                     |
| <b>BITRSHIFT</b> | Returns a number with the specified number of bits shifted to the right.                                                                                                                                                         | BITRSHIFT(<Number>, <Shift_Amount>)                           |
| <b>BITXOR</b>    | It provides a bitwise XOR of two numbers as a result.                                                                                                                                                                            | BITXOR(<number>, <number>)                                    |
| <b>COALESCE</b>  | Identifies the first expression that is not BLANK and returns it. BLANK is returned if all expressions evaluate to NULL.                                                                                                         | COALESCE(<expression>, <expression>[, <expression>]...)       |

| Function      | Description                                                                                                           | Syntax                                                                    |
|---------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| <b>FALSE</b>  | The logical value FALSE is returned by this function.                                                                 | FALSE()                                                                   |
| <b>TRUE</b>   | The logical value TRUE is returned by this function.                                                                  | TRUE()                                                                    |
| <b>NOT</b>    | This function converts either FALSE or TRUE values into the other.                                                    | NOT(<logical>)                                                            |
| <b>OR</b>     | Returns TRUE if one of the arguments is verified to be TRUE. If both arguments are false, the function returns FALSE. | OR(<logical1>,<logical2>)                                                 |
| <b>SWITCH</b> | Returns one of several potential result expressions after comparing an expression to a list of values.                | SWITCH(<expression>, <value>, <result>[, <value>, <result>]...[, <else>]) |

## Conclusion

This was a quick summary of DAX and DAX Logical Functions in Power BI. Microsoft developed a very effective tool called Power BI that assists in the formulation of crucial business decisions and promotes the company's growth. The introduction of decision-making is made possible by logical functions, which are used to combine multiple conditions together as well as to determine whether a condition is true or false. One of the most helpful DAX expressions in Power BI is the logical function.

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